# ABIGAIL MILLER

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## EDUCATION

**Stanford University** | Stanford, CA — Expected Graduation: Dec 2027
MS in Aeronautics and Astronautics – GPA: TBD/4.0

**Carnegie Mellon University** | Pittsburgh, PA — Graduated w/ Honors: May 2026
BS in Mechanical Engineering, Minor in Business Administration – GPA: 4.0/4.0
Teaching Assistant for 21-259: Calculus in Three Dimensions

## EXPERIENCE

**Rauch Engineering** – *Mechanical, Electrical, & Plumbing (MEP) Engineer* — May 2026 - present
- Designing, modeling, and drawing HVAC, electrical, and plumbing systems for commercial projects
- Highly specialized in AutoCAD MEP, Revit, and HAP. Knowledgeable in local, state, and national codes
- Created a department-wide Revit template to standardize the company's AutoCAD-to-Revit transition

**GE Aerospace** – *Thermal Systems Design (TSD) Intern* — Summer 2025
- Compiled 60+ data points into 7 flow-dependent scorecards for a DDR (Detailed Design Review)
- Analyzed and digitized measurements of the 4 core seals of 12+ engines into a single reference location
- Assessed if seal rubs meet depth & width requirements specified in our design practice
- Facilitated tours of different manufacturing/test areas at the site as chair of Lynn Intern Committee

**EMIT Lab @ CMU** – *Student Researcher* — Jan - May 2025
- Built a conductive heat model from scratch to simulate the temperature distribution of metal parts
- Optimized the density of the sample while reaching a target peak temperature (800°F)

**Westinghouse Electric Corporation** – *Fluid Systems & Procedures Intern* — Summer 2024
- Completed a sizing calculation for one of the passive cooling features
- Built a model of an active cooling system in AFT Fathom and made changes to improve flow rate

## PROJECTS

**Mini Wind Turbine Design & Fabrication** | *Renewable Energy Engineering* — February 2024
- Designed and fabricated a mini wind turbine using MATLAB, SOLIDWORKS, and a 3D printer
- Achieved a max rpm of 1.75 and 2.3V of power under 10mph wind conditions

**Planetary Rover Design & Project Management** | *NASA Mission Concept Academy* — Summer 2023
- Managed a hypothetical mission to follow a $300million budget, 6-year time constraint, & NASA Mission guidelines

## ACTIVITIES

**NASA Lunabotics Challenge** – *Structures Team Lead* — 2025 – 2026
- Required to build a lunar robot capable of performing berm construction operations
- Responsible for designing & manufacturing the chassis, spring-raised sensor mast, and subsystem integration
- Part of the 65% of teams to advance to competition; Won the Presentation & Demonstration First Steps Award

**Fringe Buggy** - *Head Mechanic ('25 & '26) and Driver* — 2023 – 2026
- Built and drove an aerodynamic motorless, man-powered racecar
- Led a team of 30+ people during builds, practices, and with machining parts
- Remodeled the CAD to fix all errors, implemented proper part structure, used CFD to analyse aerodynamics
- Created the club's first full budget + spending breakdown, and a comprehensive material procurement sheet

**Women in MechE** - *Lunch & Learn Chair* — 2023 – 2026
- Organized L&Ls for our members with both company recruiters and university professors

**Tau Beta Pi** and **Pi Tau Sigma** – *Secretary for PTS* — 2024 – 2026
- Honor societies for top 1/8 of the College of Engineering and top ¼ of the MechE class, respectively

## SKILLS & AWARDS

Skills: 3D Modeling (AutoCAD, SOLIDWORKS, OnShape, Creo Parametric, NX), ANSYS, Revit, Metal Machining, Woodworking, Composites design & fabrication, MATLAB, Java, Python, R
Awards: Northrop Grumman scholarship, Tau Beta Pi scholarship, GBU scholarship